KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3082 | 3 Credits: 3 | Program Core | Source-grounded

# Sensors and Transducers

 To introduce the basic principles of different types of sensors and transducers

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3082 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3082.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

# Official course information

### Course information

| Item | Official information |
| --- | --- |
| Programme | Diploma in Instrumentation Engineering |
| Course code | 3082 |
| Course title | Sensors and Transducers |
| Semester | 3 Credits: 3 |
| Course category | Program Core |
| Periods per week | 3 (L:2 T:1 P:0) Periods per semester: 45 |
| Periods per semester | 45 |
| Credits | 3 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

13 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

STUDY METHOD

# How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

- To introduce the basic principles of different types of sensors and transducers
- To learn different transducers and their basic applications.
- To know the basic idea about smart sensors, smart transmitters and industrial
- transmitters.

### Course outcomes

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | Identify different types of resistive transducers. 13 Applying Explain different types of inductive, capacitive | See official syllabus |
| CO2 | 10 Understanding and piezo electric transducers. Explain different types of photo electric | See official syllabus |
| CO3 | transducers, radiation detectors, ultrasonic 10 Understanding detectors, Hall effect transducers. Explain the concepts of transmitters, smart | See official syllabus |
| CO4 | sensors, smart transmitters and basic idea about 10 Understanding industrial transmitters. | See official syllabus |

### CO–PO mapping evidence

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 3 3 |
| CO2 | CO2 3 |
| CO3 | CO3 3 |
| CO4 | CO4 3 |

COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

### Module coverage

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | Identify different types of resistive transducers. | 3 | As specified |
| Module 2 | transducers. | 4 | As specified |
| Module 3 | detectors, ultrasonic detectors, Hall effect transducers. | 3 | As specified |
| Module 4 | and basic idea about industrial transmitters. | 3 | As specified |

**MODULE 1**

# Identify different types of resistive transducers.

This module is presented from the official syllabus scope for Sensors and Transducers. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Identify different types of resistive transducers.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ 5 Understanding classify them. Identify different types of resistive

5 Understanding classify them. Identify different types of resistive is an official topic in Module 1 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding classify them. Identify different types of resistive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding classify them. identify different types of resistive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
| --- | --- |

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 5 Understanding classify them. Identify different types of resistive means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 5 Understanding classify them. Identify different types of resistive □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### ▼ 5 Applying transducers and loading effect. Explain the applications of different types

5 Applying transducers and loading effect. Explain the applications of different types is an official topic in Module 1 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 5 Applying transducers and loading effect. Explain the applications of different types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 5 applying transducers and loading effect. explain the applications of different types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 5 Applying transducers and loading effect. Explain the applications of different types means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 5 Applying transducers and loading effect. Explain the applications of different types □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**▼ 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge; Explain different types of inductive, capacitive and piezo electric**

3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge; Explain different types of inductive, capacitive and piezo electric is an official topic in Module 1 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

## Study points

- Define or identify 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge; Explain different types of inductive, capacitive and piezo electric using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 3 understanding of resistive transducers. introduction to transducers, resistive transducers: define transducer and sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; resistive transducers: basic principle of resistive transducers; working of linear potentiometer, loading effect; basic principle of strain gauge, different types, derivation of equation for gauge factor, construction and working of light dependant resistor (ldr), resistance temperature detector (rtd) and thermistor; displacement measurement using potentiometer; weight measurement using strain gauge; explain different types of inductive, capacitive and piezo electric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge; Explain different types of inductive, capacitive and piezo electric means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge; Explain different types of inductive, capacitive and piezo electric □□□□□□□□□□□ □□□□□

definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# transducers.

This module is presented from the official syllabus scope for Sensors and Transducers. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: transducers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

**▼ transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of**

transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of is an official topic in Module 2 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, transducers and classification of inductive 2 understanding transducers. explain the construction and working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### ▾ LVDT and eddy current transducer and 3 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types

LVDT and eddy current transducer and 3 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types is an official topic in Module 2 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify LVDT and eddy current transducer and 3 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, lvdt and eddy current transducer and 3 understanding lvdt for pressure measurement. explain basic principle of capacitive transducers and working of different types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What LVDT and eddy current transducer and 3 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ LVDT and eddy current transducer and 3 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### ▼ 3 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric

3 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric is an official topic in Module 2 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding of capacitive transducers and capacitive level gauge. explain working principle of piezo electric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

##### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 3 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**▼ transducers and piezo electric pressure 2 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain different types of photo electric transducers, radiation**

transducers and piezo electric pressure 2 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain different types of photo electric transducers, radiation is an official topic in Module 2 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify transducers and piezo electric pressure 2 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain different types of photo electric transducers, radiation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, transducers and piezo electric pressure 2 understanding transducer. inductive, capacitive and piezo electric transducers: basic principle of inductive transducer; classification of inductive transducers; construction and working of linear variable differential transformer (lvdt); pressure measurement using lvdt ; basic principle of eddy current transducers; illustrate basic principle of capacitive transducer; working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); level measurement using capacitive transducer; basic principle of piezo electric transducer; equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; equivalent circuit of piezo electric transducer; pressure measurement using piezo electric transducer. explain different types of photo electric transducers, radiation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What transducers and piezo electric pressure 2 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain different types of photo electric transducers, radiation means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ transducers and piezo electric pressure 2 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable

Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain different types of photo electric transducers, radiation □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# detectors, ultrasonic detectors, Hall effect transducers.

This module is presented from the official syllabus scope for Sensors and Transducers. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: detectors, ultrasonic detectors, Hall effect transducers.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

**▾ photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic**

photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic is an official topic in Module 3 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, photo emissive transducers (photo 5 understanding multiplier tube), solar cell and photo voltaic cell. explain the basic principle of ultrasonic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### ▼ transducer and level measurement 2 Understanding application. Explain the construction and working

transducer and level measurement 2 Understanding application. Explain the construction and working is an official topic in Module 3 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify transducer and level measurement 2 Understanding application. Explain the construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, transducer and level measurement 2 understanding application. explain the construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

##### Topic study guide

| Aspect | What to prepare |
|---|---|

| Aspect | What to prepare |
|---|---|
| Definition/identity | What transducer and level measurement 2 Understanding application. Explain the construction and working means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ transducer and level measurement 2 Understanding application. Explain the construction and working □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

**▼ principle of Hall effect transducer and 3 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of transmitters, smart sensors, smart transmitters**

principle of Hall effect transducer and 3 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of transmitters, smart sensors, smart transmitters is an official topic in Module 3 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify principle of Hall effect transducer and 3 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of transmitters, smart sensors, smart transmitters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, principle of hall effect transducer and 3 understanding current measurement application. photoelectric, ultrasonic and hall effect transducers: illustrate photoelectric transducers- working principle of photo conductive -photo diode, photo transistor- transducers. working principle of photo emissive -photo multiplier tube- transducers, working principle of solar cell and photo voltaic cell. working principle of ultra sonic transducer. construction and working principle of hall effect transducer. application of ultrasonic transducer for level measurement, hall effect transducers for current measurement. explain the concepts of transmitters, smart sensors, smart transmitters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What principle of Hall effect transducer and 3 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of |

| Aspect | What to prepare |
|---|---|
| | photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of transmitters, smart sensors, smart transmitters means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ principle of Hall effect transducer and 3 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of transmitters, smart sensors, smart transmitters □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

# and basic idea about industrial transmitters.

This module is presented from the official syllabus scope for Sensors and Transducers. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: and basic idea about industrial transmitters.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

### ▾ Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and Smart Transmitters using basic block diagram-

Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and Smart Transmitters using basic block diagram- is an official topic in Module 4 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and Smart Transmitters using basic block diagram- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, explain the concept of transmitters 2 understanding explain the smart sensors and smart transmitters using basic block diagram- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

##### Topic study guide

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and Smart Transmitters using basic block diagram- means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and Smart Transmitters using basic block diagram- □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

### ▼ 6 Understanding Nano Sensors and pressure measurement using Nano Sensors. Explain the basic concepts about industrial

6 Understanding Nano Sensors and pressure measurement using Nano Sensors. Explain the basic concepts about industrial is an official topic in Module 4 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 6 Understanding Nano Sensors and pressure measurement using Nano Sensors. Explain the basic concepts about industrial using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 6 understanding nano sensors and pressure measurement using nano sensors. explain the basic concepts about industrial should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

##### Topic study guide

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 6 Understanding Nano Sensors and pressure measurement using Nano Sensors. Explain the basic concepts about industrial means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 6 Understanding Nano Sensors and pressure measurement using Nano Sensors. Explain the basic concepts about industrial □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**▼ 2 Understanding transmitters. Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example- Industrial transmitters-Direct mount and remote seal type transmitters- Capacitive type differential pressure transmitters. Text / Reference: T/R Book Title / Author A.K. Sawhney, "A course in Electrical and Electronic measurement and R1 Instrumentation", Dhanpathrai and Co. R2 D.V.S. Murthy, " Transducers and Instrumentation", PHI. Pvt. Ltd. D.Patranabis, " Principles of Industrial Instrumentation", Tata McGraw-Hill. R3 .Education, J.B.Gupta, "Electrical and Electronic Measurements and instrumentation", R4 Kataria Pulishers. B. C. Nakra, K. K. Chaudhry, "Instrumentation, Measurement And Analysis", R5 Tata McGraw-Hill Publishers. R6 Bela G. Liptak, "Process Measurement and Analysis", Chilton Book Company. R7 Ranganathan S, " Transducer engineering ", Allied publishers. R8 R.K. JAIN, "Mechanical and Industrial Measurements", Khanna Publishers. R9 Krishna Kant, "Computer-Based Industrial Control", PHI. R10 D. Patranabis, " Sensors and Transducers", PHI. R11 Ernest O. Doebelin, "Doebelin's Measurement Systems", McGraw-Hill. Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_ 11 VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-**

**basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E**

2 Understanding transmitters. Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example- Industrial transmitters-Direct mount and remote seal type transmitters- Capacitive type differential pressure transmitters. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, R5 Tata McGraw-Hill Publishers. R6 Bela G. Liptak, “Process Measurement and Analysis”, Chilton Book Company. R7 Ranganathan S, “ Transducer engineering ”, Allied publishers. R8 R.K. JAIN, “Mechanical and Industrial Measurements”, Khanna Publishers. R9 Krishna Kant, “Computer-Based Industrial Control”, PHI. R10 D. Patranabis, “ Sensors and Transducers”, PHI. R11 Ernest O. Doebelin, “Doebelin's Measurement Systems”, McGraw-Hill. Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/

https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_ 11 VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presentation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E is an official topic in Module 4 of Sensors and Transducers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding transmitters. Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example- Industrial transmitters-Direct mount and remote seal type transmitters- Capacitive type differential pressure transmitters. Text / Reference: T/R Book Title / Author A.K. Sawhney, "A course in Electrical and Electronic measurement and R1 Instrumentation", Dhanpathrai and Co. R2 D.V.S. Murthy, " Transducers and Instrumentation", PHI. Pvt. Ltd. D.Patranabis, " Principles of Industrial Instrumentation", Tata McGraw-Hill. R3 .Education, J.B.Gupta, "Electrical and Electronic Measurements and instrumentation", R4 Kataria Pulishers. B. C. Nakra, K. K. Chaudhry, "Instrumentation, Measurement And Analysis", R5 Tata McGraw-Hill Publishers. R6 Bela G. Liptak, "Process Measurement and Analysis", Chilton Book Company. R7 Ranganathan S, " Transducer engineering ", Allied publishers. R8 R.K. JAIN, "Mechanical and Industrial Measurements", Khanna Publishers. R9 Krishna Kant, "Computer-Based Industrial Control", PHI. R10 D. Patranabis, " Sensors and Transducers", PHI. R11 Ernest O. Doebelin, "Doebelin's Measurement Systems", McGraw-Hill. Online Resources: Sl.No Website Link

1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_ 11 VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 2 understanding transmitters. transmitters, smart sensors, smart transmitters, nano sensors: concept of standard electrical (4-20ma system) transmission; smart sensors- basic block diagram. smart transmitters- basic block diagram: nano sensor basics with example- industrial transmitters-direct mount and remote seal type transmitters- capacitive type differential pressure transmitters. text / reference: t/r book title / author a.k. sawhney, “a course in electrical and electronic measurement and r1 instrumentation”, dhanpathrai and co. r2 d.v.s. murthy, “ transducers and instrumentation”, phi. pvt. ltd. d.patranabis, “ principles of industrial instrumentation”, tata mcgraw-hill. r3 .education, j.b.gupta, “electrical and electronic measurements and instrumentation”, r4 kataria pulishers. b. c. nakra,

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### Topic study guide

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 2 Understanding transmitters. Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example-Industrial transmitters-Direct mount and remote seal type transmitters-Capacitive type differential pressure transmitters. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 |

| Aspect | What to prepare |
| --- | --- |
| | .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, R5 Tata McGraw-Hill Publishers. R6 Bela G. Liptak, “Process Measurement and Analysis”, Chilton Book Company. R7 Ranganathan S, “ Transducer engineering ”, Allied publishers. R8 R.K. JAIN, “Mechanical and Industrial Measurements”, Khanna Publishers. R9 Krishna Kant, “Computer-Based Industrial Control”, PHI. R10 D. Patranabis, “ Sensors and Transducers”, PHI. R11 Ernest O. Doebelin, “Doebelin's Measurement Systems”, McGraw-Hill. Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODlIO/TRANSDUCERS_ 11 VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presentation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 2 Understanding transmitters. Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example- Industrial transmitters-Direct mount and remote seal type transmitters- Capacitive type differential pressure transmitters. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, R5 Tata McGraw-Hill Publishers. R6 Bela G. Liptak, “Process Measurement and Analysis”, Chilton Book Company. R7 Ranganathan S, “ Transducer engineering ”, Allied publishers. R8 R.K. JAIN, “Mechanical and Industrial Measurements”, Khanna Publishers. R9 Krishna Kant, “Computer-Based Industrial Control”, PHI. R10 D. Patranabis, “ Sensors and Transducers”, PHI. R11 Ernest O. Doebelin, “Doebelin's Measurement Systems”, McGraw-Hill. Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_ 11 VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

# Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

# Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

## Module 1

▶ **Define or explain 5 Understanding classify them. Identify different types of resistive. (3 marks)**

▶ **Define or explain 5 Applying transducers and loading effect. Explain the applications of different types. (3 marks)**

▶ **Define or explain 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers; Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependant resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer; Weight measurement using strain gauge; Explain different types of inductive, capacitive and piezo electric. (3 marks)**

## Module 2

▶ **Define or explain transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of. (3 marks)**

▶ **Define or explain LVDT and eddy current transducer and 3 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types. (3 marks)**

▶ **Define or explain 3 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric. (3 marks)**

▶ **Define or explain transducers and piezo electric pressure 2 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT ; basic principle of Eddy current transducers; Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer; Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain different types of photo electric transducers, radiation. (3 marks)**

## Module 3

▶ **Define or explain photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic. (3 marks)**

▶ **Define or explain transducer and level measurement 2 Understanding application. Explain the construction and working. (3 marks)**

▶ **Define or explain principle of Hall effect transducer and 3 Understanding current measurement application. Photoelectric,**

**Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of transmitters, smart sensors, smart transmitters. (3 marks)**

## Module 4

**▶ Define or explain Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and Smart Transmitters using basic block diagram-. (3 marks)**

**▶ Define or explain 6 Understanding Nano Sensors and pressure measurement using Nano Sensors. Explain the basic concepts about industrial. (3 marks)**

**▶ Define or explain 2 Understanding transmitters. Transmitters, Smart Sensors, Smart Transmitters, Nano Sensors: Concept of standard electrical (4-20mA system) transmission; Smart sensors- basic block diagram. Smart Transmitters- basic block diagram: Nano sensor basics with example- Industrial transmitters-Direct mount and remote seal type transmitters- Capacitive type differential pressure transmitters. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. B. C. Nakra, K. K. Chaudhry, “Instrumentation, Measurement And Analysis”, R5 Tata McGraw-Hill Publishers. R6 Bela G. Liptak, “Process Measurement and Analysis”, Chilton Book Company. R7**

**Ranganathan S, " Transducer engineering ", Allied publishers. R8 R.K. JAIN, "Mechanical and Industrial Measurements", Khanna Publishers. R9 Krishna Kant, "Computer-Based Industrial Control", PHI. R10 D. Patranabis, " Sensors and Transducers", PHI. R11 Ernest O. Doebelin, "Doebelin's Measurement Systems", McGraw-Hill. Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODlIO/TRANSDUCERS_ 11 VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presentation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E. (3 marks)**

PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **5 Understanding classify them. Identify different types of resistive**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **transducers and classification of inductive 2 Understanding transducers. Explain the construction and working of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and**

## Module 4

1-mark definition: State the meaning of **Explain the concept of transmitters 2 Understanding Explain the Smart Sensors and**

**photo voltaic cell. Explain the basic principle of Ultrasonic**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

**Smart Transmitters using basic block diagram-**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

# Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

# Glossary

### Study glossary

| Term | Meaning in this lesson |
|---|---|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## Web resources listed in the official PDF


- Sl.No Website Link https://www.youtube.com/watch?v=FYggY7LLDrw
- https://www.slideshare.net/manash234/classification-of-transducers
- https://www.electronics-tutorials.ws/io/io_1.html
- https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf https://www.youtube.com/watch?v=XSbUnUADlno
- http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html https://www.youtube.com/watch?v=anCnrtjNLQM
- http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-tutorial.pdf
- https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html
- https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers-
- https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_
- VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen
- https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ https://www.youtube.com/watch?v=CckMErCuXvE https://www.youtube.com/watch?v=7jQ12uUqK6E


IN-PAGE SEARCH

# Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3082. Verify institutional updates against the current official curriculum.